

Tandem multipass Fabry-Perot interferometer for Brillouin scattering

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A new design of a plane mirror Fabry-Perot interferometer with the mirror blanks freely supported by horizontal rings is described. Special attention is paid to the surface flatness and dielectric coatings of the mirrors. It is shown that any two piezoelectrically scanned Fabry-Perot interferometers can be synchronized in tandem operation using simple optics and standard electronic driving and stabilization equipment. Applications of the tandem system to Brillouin scattering are given. The observation of Brillouin spectra of supported thin plastic films and of a clamped metal surface is reported.

I. Introduction

In 1961, Hariharan and Sen¹ showed that the contrast and finesse of a plane mirror Fabry-Perot interferometer are significantly increased by routing the beam sequentially through two different portions of the mirrors using a cat's-eye reflector. This highly sharpened filter was attained at the relatively low cost of a decrease in overall transmission and throughput.

Nine years later, Sandercock² applied this idea and introduced a more general form of a multipass interferometer. He used corner-cube prisms to route the beam three to five times through different areas of the mirrors. In this way, a contrast in excess of 10^9 was achieved as was necessary for looking at weak signals from opaque surfaces, which cause a tremendous amount of stray scattering.

At the same time, several forms of tandem interferometers appeared. Pine³ constructed a spectrometer consisting of a plane mirror Fabry-Perot interferometer in series with a confocal one. He used the plane mirror interferometer as a fixed filter to reject stray scattering and scanned the spherical one for resolving the spectrum of interest. Vacher and co-workers have developed a similar instrument, a new refinement of which

was published recently in this journal.⁴

Cannell^{5,6} took a different approach to the tandem system. He showed that when two pressure-scanned Fabry-Perot interferometers were placed in series in the same pressure tank, the contrast and finesse of the resulting filter were not only improved, but also a significant increase in the free spectral range was obtained. This latter advantage alleviates the problem of overlapping orders. The synchronously scanned tandem interferometer has therefore an interesting and useful instrumental profile. If it had not been for the seemingly difficult electronic problems, a piezoelectrically scanned tandem system would have been built at the beginning of the seventies.

Sandercock⁷ recently introduced an elegant design for a piezoelectrically scanned tandem multipass Fabry-Perot interferometer. His design makes use of a mechanical scan where one mirror of each interferometer is mounted on one and the same base plate, which in turn is scanned by one piezostack. The two mirror pairs are set at an angle which determines the mirror spacing ratio of the two interferometers.

Independent of Sandercock's work, we have taken a different approach to the synchronous scan of a piezoelectrically driven tandem system. It is the purpose of the present paper to show that with the aid of some simple optics, any two Fabry-Perot interferometers can be scanned synchronously using standard electronic driving and stabilization circuitry. We also describe a new design of a plane mirror Fabry-Perot interferometer. In this design, the mirror blanks are freely supported by horizontal rings so that the problem of stress-free mirror mounting is virtually eliminated, and interchanging of mirror pairs is almost trivial. Finally,

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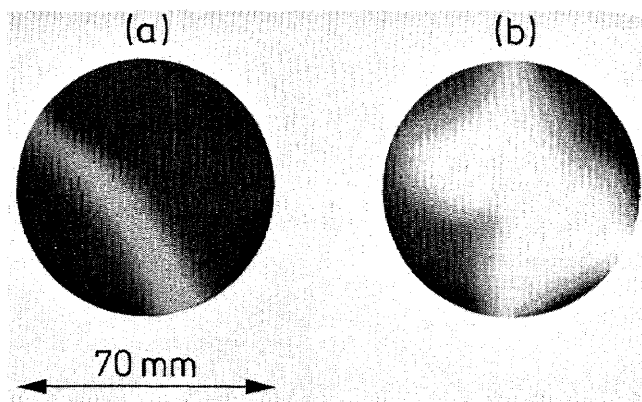


Fig. 1. Interferograms of a good pair of 70-mm diam mirrors: fringes are (a) 0.5 m apart and (b) with infinite fringe spacing.

we give several applications of the tandem system to Brillouin spectroscopy.

II. Single-Pass and Multipass Interferometers

We begin this section by considering the single-pass transmission function of a plane mirror Fabry-Perot interferometer:

$$A(k, d, i) = \frac{T_1}{1 + \frac{4N_1^2}{\pi^2} \sin^2[kd \cos(i)]}, \quad (1)$$

where T_1 is the overall single-pass transmission, N_1 is the effective single-pass finesse, d is the mirror separation, and i is the angle of incidence of the illuminating light beam of wave number

$$k \equiv 2\pi n/\lambda. \quad (2)$$

In Eq. (2), λ denotes the free-space wavelength, and n is the refractive index of the medium between the mirror blanks ($n = 1$).

The effective finesse N_1 contains contributions from the reflective finesse and a finesse due to surface aberrations and roughness of the mirrors.⁸ It is therefore important to coat and mount the mirrors so that no surface deformation occurs.

Our 70-mm diam mirror pairs were purchased from ICOS in London. Between subsequent stages of polishing, such interferometer mirrors are silver-coated and tested horizontally in a Fizeau interferometer. During these test, each mirror rests on three pieces of aluminum foil, which in turn lie symmetrically inside a ring serving as the mirror holder. Polishing is stopped when the silver-coated mirror pair is matched better than $\lambda/200$ ($\lambda \approx 500$ nm). Their separate flatness is then typically of the order of $\lambda/50$.

The next step is to obtain a dielectric reflection coating on the mirrors with minimum absorption and scattering while not affecting the surface figure. All coatings have some kind of surface tension and tend to deform even relatively thick pieces of glass. For example, a stack of ZnS-SiO₂ layers deformed our $\lambda/200$ matched mirror pair to a poor $\lambda/50$, although the

Spectrasil blanks were 18 mm thick. It is a happy coincidence of nature that the soft dielectric coating of ZnS-MgF₂ layers both has very good absorption and scattering properties and does not affect the mirror flatness.

Our mirrors have a reflectivity of $R = 0.94 \pm 0.05$ at 500 nm. When mounted horizontally as described above, but with one mirror adjustable by means of three piezostacks, they can be set parallel and on resonance ($d = p\lambda_0/2$, p is an integer). In that position, for a perpendicularly incident ($i = 0$) parallel beam of monochromatic light, the overall transmission is typically $T_1 = 0.75 \pm 0.05$. When illuminated over their total aperture with the expanded laser beam, interferograms like the ones shown in Fig. 1 are obtained. A very good pair should light up completely, so that the areas where the beam is routed for multipassing can be chosen freely. The pair in Fig. 1 has been in use for more than 3.5 yr and still shows no sign of degradation.

There are several experimental pitfalls where the mirrors apparently light up over the whole aperture even though they may suffer from severe aberrations. Therefore, two extra tests should be performed.

First, when measuring the single-pass transmission, one should connect the detector output to an oscilloscope. Mechanical vibrations, mostly in the 1-Hz to 2-kHz region, show up as amplitude modulations in the transmitted light. These modulations may be due to a voltage ripple on the piezostacks (50 or 60 Hz), mechanical resonances of the interferometer, or laser frequency jitter. They may be as deep as 0.99, thus severely reducing the transmission and finesse of the interferometer. A mechanically rigid interferometer with a good set of mirrors of $R = 0.94$ shows <0.10 modulation in its transmitted light when the mirrors are set to half resonance [$d = \lambda_0/4N_1 + p\lambda_0/2$ in Eq. (1)], and d is so small that laser jitter cannot be resolved. A few scans per second do not bring the freely supported mirrors into vibration, but the flyback after each scan should be smooth and take ~ 0.1 sec. The single-pass finesse then takes the value $N_1 = 47 \pm 3$. We usually place the interferometer on rubber blocks for further vibration isolation.

The second test which should be done is the investigation of the illuminating wave front with, for example, a shear plate. The acceptance angle δi of the interferometer is easily calculated from Eq. (1) and takes the form

$$\delta i = \left(\frac{\lambda}{2dN_1} \right)^{1/2}, \quad (3)$$

so that all rays incident on the mirrors should fall within this angle. When the lenses in the beam expander suffer from aberrations, or when the laser oscillates in a transverse mode, all incident rays may not fall within δi , and the mirrors will light up more completely than they should.

From the foregoing it is a natural continuation to construct a Fabry-Perot interferometer with the mirrors in a horizontal position. In Ref. 9 it was reported that

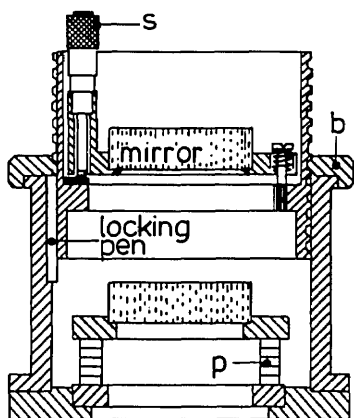


Fig. 2. Design of a Fabry-Perot interferometer with the mirror pair supported by horizontal rings: *s*, microspindles for mechanical tilt adjustment; *p*, piezostacks; *b*, nut for varying mirror spacing.

we set the commercially available Burleigh interferometer in an upright position so that the mirrors could be supported horizontally on aluminum rings. Although almost any mechanically rigid construction suffices, the tubelike geometry of Fig. 2 has some distinct advantages. This cylindrical construction is very rigid and made almost completely of aluminum. It encloses the air between the mirror blanks so that turbulence is not likely to occur. The mirror spacing is variable from 1 to 50 mm by conveniently rotating the nut *b*. For smaller mirror spacings, the upper mirror should be somewhat larger in diameter. No attempt has been made to maintain parallelism between the mirrors when their spacing is changed, since coarse mechanical adjustment of the upper mirror is easily attained by means of the two microspindles *s*.

The masks with corner-cube prisms for multipassing are easily mounted on top of and underneath the interferometer. The resulting instrumental profile is the function of Eq. (1) to the *m*th power, where *m* is the number of passes.⁸ Measurements often show that the contrast of an *m*-pass interferometer does not increase as

$$C_m = \left(1 + \frac{4N_1^2}{\pi^2}\right)^m, \quad (4)$$

but is in practice several orders of magnitude smaller. It may then be argued that the cavity formed by the corner-cube prisms and the mirrors is responsible for this effect as it may pull extra light through the system. We have never observed such resonances, and they are not likely to occur in a significant way since the acceptance angle and the finesse of this large cavity are small. Moreover, even if these resonances existed, they would easily be observed by pushing slightly against a corner-cube prism.

However, we have observed ghosts which do affect the contrast significantly. In most cases, these ghosts could be identified with resonances between the back surfaces of the mirror blanks. The reasons these ghosts appear despite accurate spatial filtering is that the mirror wedge is often chosen too close to the acceptance angle

given by Eq. (3). This wedge should be ~ 0.02 rad to prevent scattering from the back mirror surfaces from entering the last pinhole. In many practical cases, it is an order of magnitude smaller so that ghosts may appear.

III. Tandem System

When two plane mirror Fabry-Perot interferometers are placed in series, the resulting instrumental profile is given by (*i* = 0)

$$A(k, d, d') = \frac{T_1}{1 + \frac{4N_1^2}{\pi^2} \sin^2(kd)} \cdot \frac{T'_1}{1 + \frac{4(N'_1)^2}{\pi^2} \sin^2(kd')}, \quad (5)$$

where the primes indicate the corresponding parameters for the second interferometer. Suppose now that both interferometers are at resonance for k_0 at orders *p* and *p'*, respectively:

$$k_0 d = p\pi, \quad k_0 d' = p'\pi, \quad (6)$$

where *p*, *p'* are integers.

When the first interferometer is scanned over a wavelength shift $\Delta\lambda$,

$$\Delta\lambda = 2\Delta d/p, \quad (7)$$

where Δd is the mirror displacement from the original spacing *d* of the first interferometer, the second interferometer must transmit the same wavelength $\lambda_0 + \Delta\lambda$ so that

$$\Delta\lambda = 2\Delta d'/p'. \quad (8)$$

Equations (6)–(8) yield for the relative scan amplitudes

$$(\Delta d')/(\Delta d) = d'/d. \quad (9)$$

When Eqs. (6)–(9) are satisfied, the instrumental profile given by Eq. (5) has the shape shown in Fig. 3, which is equivalent to the one given in Cannell's work.⁶ In this figure, we assumed the following practical values for the parameters $T_1 = T'_1 = 0.75$, $N_1 = N'_1 = 50$, $d'/d = 0.9$. The resulting transmission function is not only sharper and has a much higher contrast than a single interferometer but also has a significantly larger free spectral range, a factor of 10 larger in the case shown. The

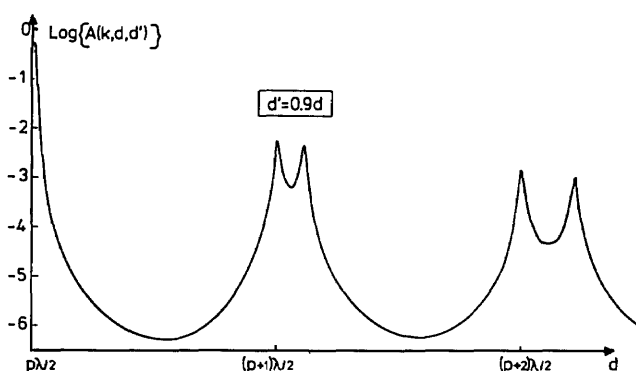


Fig. 3. Instrumental profile of a single-pass tandem interferometer with mirror spacing ratio $d'/d = 0.9$, overall transmission of each interferometer $T_1 = T'_1 = 0.75$, and effective finesse $N_1 = N'_1 = 50$.

consequence of this latter feature is that broad Stokes and anti-Stokes components will no longer overlap. When spectra are taken near a solid-liquid interface as in Ref. 9, Brillouin peaks both from the solid and from the liquid appear. The frequency shifts of the peaks from the solid are about three times larger than those from the liquid and therefore show up in higher order and tend to confuse the spectra. This will not happen when such spectra are taken with a tandem system.

The practical realization of the electronic synchronization of the two interferometers is quite straightforward. Equation (6) merely expresses the fact that the two interferometers should be locked on the same wavelength, most conveniently the laser line. During subsequent scans, these resonances of both interferometers should occur at exactly the same moment relative to the beginning of each scan. This implies that both interferometers are scanned with a ramp voltage which originates from the same clock. Electronically this means that we start from one low voltage ramp, amplify this in two separate high-voltage amplifiers, which drive the two interferometers, adjust the relative amplification of these amplifiers according to Eq. (9), and make sure that the laser line resonances of Eq. (6) occur at the same time relative to the beginning of each scan. Before we show that this electronic synchronization is easily accomplished with standard equipment, we first describe the Brillouin spectrometer with the tandem interferometer.

Consider the geometry of the spectrometer shown in Fig. 4. The 514.5-nm line of an argon laser is almost completely reflected by the beam splitter B_1 toward the sample. A small part of the transmitted beam is directed by the beam splitter B_2 to the spectrum analyzer SA for mode analysis. The beam transmitted by B_2 is then, via mirror M_2 , expanded by the lenses l_3 and l_4 to a parallel beam with a diameter of ~ 15 mm. This beam, which serves for the alignment of both interferometers, can be blocked by the shutter Sh_1 . This shutter is just a small piece of metal mounted on the spindle of a fast stepping motor.

The expanded alignment beam is directed via the beam splitters B_3 and B_4 , each of which has a reflectivity of 0.01, through the spatial filter which consists of the lenses l_5, l_6 and the pinhole p_1 . The diameter of this pinhole is given by the product of the acceptance angle of the interferometer as expressed by Eq. (3) and the focal length of the lens l_6 . The diameter of the beam of parallel light leaving this lens determines the size S of the areas selected for multipassing on the interferometer mirrors. This diameter is usually of the order of 10 mm when corner-cube prisms are used but may be chosen larger when cat's-eye reflectors are employed.

Although two mirrors, M_3 and M'_3 , are shown in Fig. 4 for convenience, only M_3 is necessary for directing the beam vertically up to the first interferometer FP_1 . This interferometer is operated in triple pass by means of the corner-cube prisms C_1 and C_2 . The beam leaving the interferometer is reflected by the mirror M_4 toward the beam splitter B_5 , which reflects a small part, say 0.01 to the photodiode D .

The parallel beam of light transmitted by the beam splitter B_5 encounters the same interferometer optics as the beam reflected by B_4 . When the alignment beam leaves the second triple-pass interferometer FP_2 , it is directed via the mirror M_6 toward the reflection grating Gr . This grating is used in Littrow mount and serves to reject the Raman spectra by means of the last spatial filter. This filter consists of the lenses l_9, l_{10} and the pinhole p_3 . It also blocks possible ghosts as described in the last section. The lens l_{10} finally images the light onto the sensitive area of the photomultiplier PM , the electrical signal of which is amplified by standard Princeton Applied Research (PAR) equipment.

The digital pulses leaving this equipment are fed into the Burleigh DAS1, which is used in combination with the high voltage amplifier HV_2 to scan and servocontrol the second interferometer FP_2 . It also accumulates the data in its multichannel analyzer. This data acquisition and stabilization system (DAS) has been described in a recent paper¹⁰ in this journal. It has the basic property that one can program a window in the ramp at a given time interval during each scan. A resonance line of the interferometer is locked in the center of this window and optimized in height by applying test and correction voltages to the piezostacks during each scan. The DAS1 has a low voltage ramp output which is amplified by the separate high-voltage module HV_2 which drives the interferometer. By varying the amplification of HV_2 , the interferometer is programmed to have a tilt-free and linear scan over as many as 6 orders. The linearity is programmed to be better than 0.004, which is sufficient for tandem operation.

The low voltage ramp output of the DAS1 is also fed into another high voltage module HV_1 , which drives the first interferometer FP_1 . The laser resonance peak

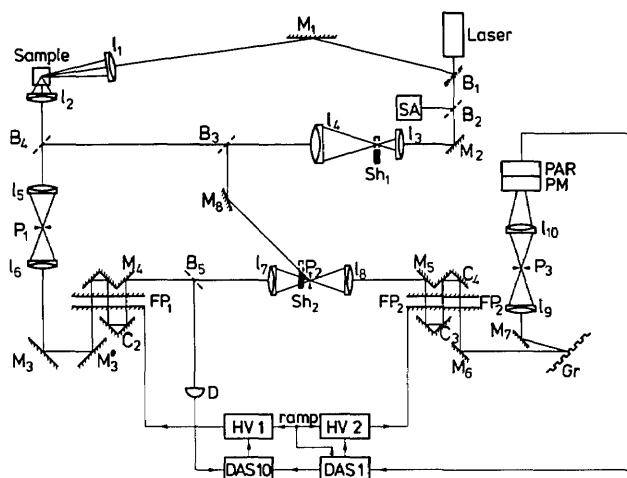


Fig. 4. Brillouin spectrometer with two Fabry-Perot interferometers $FP_{1,2}$ piezoelectrically scanned in tandem: l_1 - l_{10} , lenses; B_{1-5} , beam splitters; M_{1-8} , mirrors; p_{1-5} pinholes; C_{1-4} corner-cube prisms; D , photodiode; Gr , grating; SA , spectrum analyzer; PM , photomultiplier; PAR, Princeton Applied Research photon counting equipment; DAS, data acquisition and stabilization system; and $HV_{1,2}$, high voltage amplifiers.

from FP_1 is detected by the photodiode D whose electrical signal is fed into the Burleigh DAS10. The DAS10 in combination with HV_1 servocontrols the first interferometer. It receives a blanking pulse from DAS1 after each scan, and its optimization window is also triggered by the corresponding window of DAS1. Thus Eq. (6) is satisfied.

The alignment of the tandem system proceeds as follows. With the shutter Sh_1 in the open position, the first interferometer is brought into alignment using the clearly visible alignment beam which passes through the first spatial filter. Then this interferometer is scanned and locked into the window of DAS10. The beam leaving the first interferometer and transmitted by the beam splitter B_5 is blocked by the shutter Sh_2 , thus completely isolating the first interferometer from the second. The preliminary alignment of the second interferometer FP_2 with the following optics is established by setting the shutter Sh_2 in the open position and using the beam transmitted by FP_1 . After this preliminary alignment, Sh_2 is closed to block the light coming from FP_1 but making the way free for the light derived from the expanded laser beam by means of the beam splitter B_3 and the mirror M_8 . This light overflows the pinhole p_2 and is still strong enough to be detected by the photomultiplier so that it can serve as an alignment signal for the DAS1. The amplification of HV_2 is then turned up from zero and the resonance peak from FP_2 locked in the window of DAS1. The proper adjustment of the relative ramp amplitudes to satisfy Eq. (9) is described later in this section. Finally, the shutters Sh_1 and Sh_2 are triggered by the optimizing window so that only during locking the alignment beam is present and the two interferometers are optically isolated. We have incorporated in the DAS1 an extra feature enabling it to be programmed to activate the shutters during any time interval of each scan, including flyback.

If we had locked on the resulting resonance peak of the two interferometers as Sandercock⁷ does, the optimizing cycle of the test and correction voltages would have become rather long. In that case, an extra differential drift step would have to be included in the finesse optimization.

Our setup is flexible; the free spectral ranges of the two interferometers may be set quite arbitrarily since synchronization is guaranteed electronically. The alignment of both interferometers is established independently as is locking to the laser line. We employ two triple-pass interferometers in tandem as this is a good compromise in terms of contrast, finesse, and overall transmission.

One may argue that the three spatial filters employed are cumbersome and introduce optical losses. This is not true. They are helpful in aligning the spectrometer with the clearly visible alignment beam. Once these lenses and pinholes are set into position, they should not be touched again. In our setup, all lenses are standard achromatic doublets of 100-mm focal length. The diameter of the pinholes is then 100 μm , which allows for interferometer spacings up to 10 mm [Eq. (3)]. The overall transmission of the spectrometer from l_5 to l_{10}

is ~ 0.10 , which includes the 0.60 grating efficiency. The total setup starting from mirror M_2 and including the photomultiplier can be mounted on a base plate of 0.25 m^2 .

The alignment of the sample, the focusing, and the collecting lenses $l_{1,2}$ with the spectrometer may now proceed while the spectrometer is scanning and locked on the laser line. The axial alignment of the lens l_2 is established by placing a diffuse scattering surface at the common focus of l_1 and l_2 and causing the scattered beam collected by l_2 to pass without vignetting through the first spatial filter. After this axial alignment, the lens l_2 should only be allowed to move along the optical axis. For optimum collection efficiency, the focal lengths of l_1 and l_2 should be chosen so that the focal area S_f seen from l_2 with a solid angle $d\Omega$ satisfies the relation of constant throughput:

$$\frac{\pi S}{4} (\delta i)^2 = S_f d\Omega. \quad (10)$$

In Eq. (10), as before, δi is the acceptance angle of the interferometer, and S is the surface area of the parallel beam of light incident on the interferometer mirrors.

As a test sample, a strong Brillouin scatterer is used, for example, ethanol. With the aid of a microscope pointed at the pinhole p_1 , the focal region in the sample is focused onto this pinhole. This light mainly contains the two Brillouin components. The proper adjustment of the relative ramp amplitudes is readily achieved by maximizing the Brillouin peaks appearing on the multichannel analyzer during each scan. Both conditions of Eqs. (6) and (9) are now satisfied, and the spectrometer is ready for operation. When the temperature in the laboratory does not change more than a few degrees, the DAS1 and DAS10 keep the interferometers in tune over any desired measuring time.

Although it is not necessary to perform a calibration measurement after each change of mirror spacings or before a new series of measurements, it is useful and far from time consuming to do so. Such a measurement also yields the absolute mirror spacings of both interferometers. A change in either one can be measured mechanically very accurately. The spacing ratio is then known in advance, and the corresponding relative ramp amplitudes can be adjusted by looking at the laser resonances on the multichannel analyzer.

IV. Applications

As explained in the last section, a tandem Fabry-Perot interferometer becomes a powerful optical filter whenever the problem of overlapping orders arises. Although sophisticated numerical techniques have been developed to deconvolute the instrumental profile from the spectrogram, this deconvolution does not necessarily give a unique answer. In many cases, the very sharp filter described in the last section makes numerical deconvolution obsolete.

Overlapping orders are, for example, present in the Brillouin spectra taken from viscous polymeric liquids or soft plastics. In addition to the normal Brillouin peaks, many of these substances given spectra with an extra central component which may be wider than the Brillouin shifts. For example, Wang and co-workers¹¹

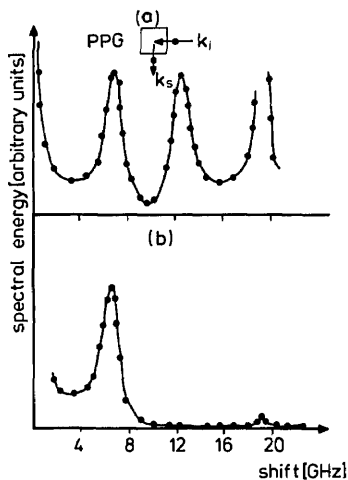


Fig. 5. Brillouin spectrum from PPG taken with (a) a single interferometer and (b) the tandem system with mirror spacing ratio $d'/d \approx 0.8$.

have done beautiful experimental and theoretical work on the structural relaxation in polypropylene glycols (PPG). Although they carefully selected the free spectral ranges for their experiments, the problem of overlapping Stokes and anti-Stokes components somewhat limited their data. We have illustrated this overlapping in Fig. 5 where the upper part shows a spectrum of PPG taken with the second triple-pass interferometer alone (mirrors removed from FP_1), whereas the lower part was taken with the tandem system. The scattering geometry is shown in the inset: k_i and k_s are the wave vectors of the incident and scattered light. The polarization is indicated by a dot or an arrow, depending on whether the light is polarized perpendicular to or in the scattering plane. Note the small ghost at 19.22 GHz, which is due to the strong stray scattering of the liquid. This ghost is not sufficiently suppressed by the first interferometer. The second interferometer has a somewhat larger contrast so that the ghost at 16 GHz ($d' \approx 0.8d$) is completely suppressed. It should be emphasized that the spectra of Fig. 5 only serve to illustrate the higher resolution in combination with the larger dynamic range of the tandem system.

Our thinking about tandem systems was initiated by our measurements near a liquid-solid interface as described in Ref. 9. In these experiments, laser light was focused near a liquid-glass interface so that Brillouin peaks of the liquid as well as of the glass were observed. Only the Brillouin spectrum of the liquid was of interest. The larger shifts corresponding to the transverse and longitudinal Brillouin doublets of the solid appeared in higher order and tended to confuse the spectrum of interest. By careful setting of the mirror spacing of the single interferometer we used, most of these doublets could be arranged to fall under the laser resonance peaks and in the center of the free spectral range. However, for some scattering geometries it was not possible to separate the peaks of the substrate from the spectrum of interest. This is a repeating problem in light-scattering experiments performed in the vicinity of interfaces.

A relatively new problem of industrial importance is the adhesion and stiffness of thin plastic films on hard glass substrates. The adhesion of such a film onto its substrate can be investigated by means of the Brillouin scattering technique described in Ref. 9. The hypersound stiffness is directly calculated from the frequency shifts of the Brillouin peaks. Therefore, Brillouin scattering is a unique tool to investigate the elastic properties of such supported thin films, and we shall see that here lies a powerful application of the tandem system.

In this paper we want to show an interesting feature of supported thin plastic films. These films act as acoustic waveguides for the thermal acoustic phonons in complete analogy with the planar optical waveguides in integrated optics. However, unlike the electromagnetic waves, the acoustic modes can have transverse as well as longitudinal polarization. The transverse modes polarized in the scattering plane are coupled to the longitudinal ones by means of the acoustic boundary conditions at the interfaces. These so-called generalized Lamb modes¹² are observed in the backscattering geometry illustrated in the upper spectrum of Fig. 6. In this geometry, the transverse acoustic phonons polar-

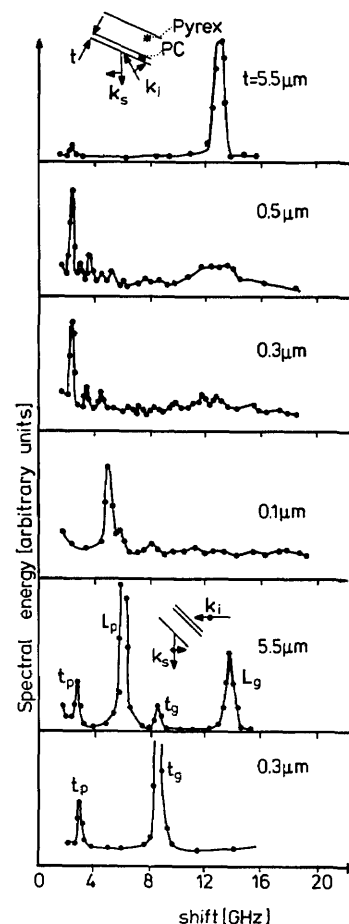


Fig. 6. Brillouin spectra of polycarbonate films of varying thicknesses on Pyrex substrates taken with the tandem system; the scattering geometries are shown in the insets.

ized perpendicular to the scattering plane should not be observed according to the standard Brillouin scattering selection rules.

We have used polycarbonate (PC) films supported by Pyrex substrates. From a light-scattering point of view, these films have the advantage that their optical contrast with the substrate is negligible ($n_{\text{Pyrex}} = 1.47$ and $n_{\text{PC}} = 1.59$), whereas the acoustic contrast, which becomes apparent in a Brillouin scattering experiment, is significant.

Consider now Fig. 6. When the PC film is $5.5 \mu\text{m}$ thick, the zero-order transverse Lamb mode appears already in the spectrum as the small peak at 2.40 GHz. The strong peak at 13.15 GHz corresponds to the longitudinal Brillouin peak of bulk PC. With decreasing film thickness, the discrete acoustic modes become apparent and more separated. Especially the narrow transverse modes with the lower shifts are most distinct. It is interesting to note that a Brillouin spectrum is readily observed, even from a film $0.1 \mu\text{m}$ thick. The data accumulation time of this spectrum is 30 min with an incident laser power of 0.5 W.

The last two spectra of Fig. 6 were taken in a forward scattering geometry (see inset) so that the transverse modes polarized perpendicular to the scattering plane should be observed. The first of this series shows the bulk spectra of both the film and the Pyrex substrate. In this symmetric scattering geometry, the frequency shifts ν of the transverse and longitudinal peaks, (t_p, t_g) and (L_p, L_g), of the plastic and glass, respectively, directly yield the transverse and longitudinal hypersound velocities $v = 2\pi\nu/\sqrt{2}k_i$. The two transverse peaks are now depolarized. With decreasing film thickness, a discrete set of these depolarized peaks due to the so-called Love modes¹² in the polycarbonate films should be observable. As the last spectrum of Fig. 6 shows, only the zero-order depolarized mode is present. When this spectrum was taken, a polaroid was placed in the direction of observation to block the polarized scattering from Lamb modes.

As a last application we discuss some new measurements on liquid metals. In our former paper,⁹ we proposed a scattering experiment where a Brillouin spectrum is taken from liquid mercury resting against a glass substrate. The mercury surface is then nearly clamped so that no scattering from the stress-free surface ripples as in our early work on liquid metals¹³ is allowed to occur. Figure 7 shows the relevant Brillouin spectrum with the scattering geometry shown in the inset. The incident and scattered light are polarized in the scattering plane. The data are accumulated for 2 h with 0.3-W incident power. This spectrum has the following new features. The strong peak next to the laser line is due to the scattering from the combined surface ripples of the glass substrate and the mercury. The frequencies of these so-called Stoneley waves^{12,14} have about half of the values of the corresponding frequencies of the Rayleigh waves on the stress-free glass surface and are ~ 1 -2 orders of magnitude higher than those of the ripplons on a free liquid mercury surface. The shift of this peak varies with $\cos(\theta_i) + \cos(\theta_s)$ [θ_i and θ_s are the

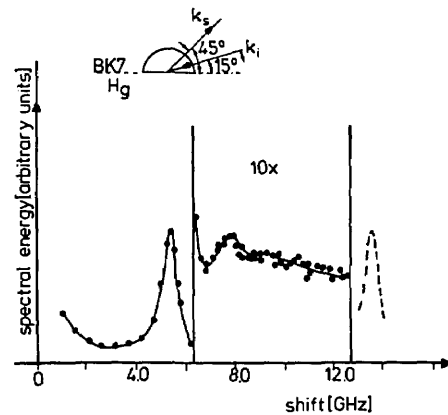


Fig. 7. Brillouin spectrum taken at a clamped liquid mercury surface; scattering geometry is given in the inset. The sharp peak close to the laser line is due to the Stoneley surface phonons at the glass-mercury interface. The small jump at the tail of this peak is due to the bulk phonons in the mercury. The peak of the broken line is observed when instead of the mercury the back surface of the glass is coated with aluminum.

angles that $\mathbf{k}_i$ and $\mathbf{k}_s$ make with the interface (Fig. 7)] and is slightly dependent on the elastic constants of the different glasses we used as substrates (fused silica, BK7, SF59, and gadolinium gallium garnet). As Fig. 7 shows, it rests on a central component whose origin is presumably due to the isobaric density fluctuations in the mercury. On the tail of the surface peaks, one clearly discerns a discontinuous jump in the data. This jump corresponds to the cutoff frequency of the Brillouin spectrum of a clamped liquid mercury surface as predicted in Ref. 9. The tail of this latter spectrum decreases in intensity because it rests on a weak and diminishing central component due to the glass substrate.

We have also taken spectra from the glass demispheres with their back surfaces coated with a 40-nm thick aluminum layer. The resulting shifts originating from the Stoneley surface waves are a factor of 2.5 larger than in the case of liquid mercury (broken line in Fig. 7).

This concludes the section on applications of the tandem Fabry-Perot interferometer to Brillouin scattering. For further applications to surface phonon scattering on opaque solids and spin wave scattering, the reader is referred to Sandercock's work.⁷

V. Conclusions

We have shown that the construction of a multipass Fabry-Perot interferometer is quite simple when the mirror blanks are supported by horizontal rings. When the mirrors are coated with a stack of ZnS-MgF₂ layers, the mirror flatness is not affected, and only low scattering and absorption losses are present.

It is also shown that the operation of any two Fabry-Perot interferometers piezoelectrically scanned in

tandem is not difficult with standard electronic scanning and stabilization equipment when some simple optical alignment procedures are followed. The main ideas behind tandem synchronization are that both interferometers are scanned with a ramp originating from one clock and that the interferometers are optically isolated during the small time interval of locking to the laser line.

Applications of the tandem system to Brillouin scattering are given. The extra feature of a tandem interferometer is its relatively large free spectral range resulting in less overlap of orders. Full merit of this system is obtained when broad spectra are to be observed or when a range of peaks over a wide frequency regime is expected. The system has potential uses for research areas in polymer science.

Finally, we have reported the observation of a Brillouin spectrum from a clamped liquid-metal surface, a spectrum which we were not able to resolve with a single interferometer. This experiment also yielded a simple scattering geometry for observing the surface phonons of the Stoneley type.

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Electromagnetic Theory of Gratings. Edited by R. PETIT. Springer-Verlag, Heidelberg, 1980. 284 pp. \$38.35.

Although diffraction gratings were invented in the eighteenth century by David Rittenhouse and improved over the years by various scientists, very little was done toward calculating their efficiencies. During the last decade and a half, steps have been taken to fill this hiatus by a group at the University of Aix-Marseille III, headed by R. Petit. This is not to say that Petit and his group were the first, or are the only ones, to calculate efficiencies, but they have mounted a concerted effort to solve the problem that has met with much success—and they have written the first book on the subject. This book can be considered as an overview of the work done to date, in calculating efficiencies of plane gratings. Rigorous calculations for concave gratings are still beyond the state of the art. Most of the contents are by Petit and his colleagues, but the final chapter was written by three residents of Australia, a country where there is also a group working on the problem of grating efficiencies.

The book contains seven chapters: a *tutorial introduction* by Petit; *some mathematical aspects of the grating theory* by M. Cadilhac; *integral methods* by D. Maystre, *differential methods* by P. Vincent; *the homogeneous problem* by M. Neviere; *experimental verifications and applications of the theory* by Maystre, Neviere, and Petit; and *theory of crossed gratings* by R. C. McPhedran, G. H. Derrick, and L. C. Botten.

The first two chapters, *tutorial introduction* and *some mathematical aspects . . .*, provide an explanation of how the problem is reduced to mathematics and some of the techniques required to obtain the solutions. It should be mentioned here that in this book the orientation of the electric vector is with respect to the grooves; $E_{\parallel}$ is parallel to the grooves and $E_{\perp}$ is perpendicular to them. The usual orientation convention, e.g., as used in calculating reflectance using Fresnel equations, is to refer the direction of the electric vector to the plane of incidence so that $E_{\perp}$, perpendicular to the plane of incidence, would be parallel to the grooves when the grating is used in the conventional manner.

Integral methods (IM) shows how to reduce the problem of grating diffraction to the solution of a linear integral equation or a system of coupled linear integral equations. This technique appears to be the most general for a rigorous solution of the problem, although not the simplest for computing or for providing an intuitive understanding. The *differential method* (DM) can also be used in rigorous calculations of grating efficiencies by the solution of partial differential equations with suitable boundary conditions. However, the general conditions are more restrictive than for the IM. The IM is capable of dealing with materials having very high conductivity, but apparently large conductivities cause numerical problems in the algorithms used in computing with the DM. In fact, the DM will not work for perfectly conducting surfaces, and it is necessary to resort to a conformal mapping method that is competitive with the IM for gratings with these surfaces. One might ask, why bother with perfectly conducting surfaces since there are, in fact, none. The authors point out that the properties of gratings with metal surfaces can be calculated quite accurately assuming perfectly conducting surfaces for wavelengths $>4 \mu\text{m}$.

The homogeneous problem deals with grating anomalies; phenomena connected with the excitation of surface waves along a periodic structure. This section also includes grating couplers used in integrated optics and a theory of the grating coupler is presented. One of the unexpected results arising from this study is the fact that a grating with a highly reflecting coating can be made to absorb the incident wave completely if the ratio of grating spacing to groove depth is adjusted properly.

To obtain *experimental verifications . . .* of the calculations, one must measure the efficiency of the grating in all possible orders, preferably with polarized radiation, the groove profile must be de-

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